Natural Language Technology

Week 4 - Language Output Technology

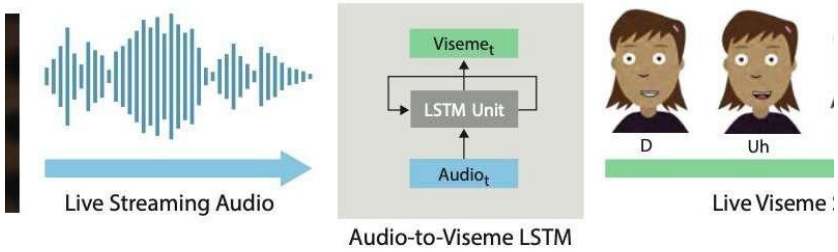
Recap Week 2

- What is language?
- The study of language
- Optical Character Recognition
- Handwriting Recognition
- Speaker Recognition
- Speech Recognition
- Saving audio
- Using Google Speech to Text v2 through API
- Estimate word error rate for various inputs

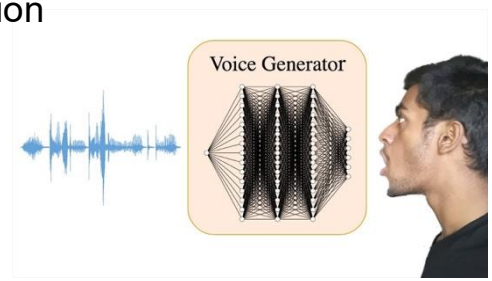
Today: Language Output Technologies



Text to Speech (TTS)



Lip Sync Character Animation



Voice Cloning

Today

- Speech Synthesis
  - What are the properties of Language which have made it challenging to synthesise?
- Text to Speech Systems
- Voice Cloning
- Visual Outputs

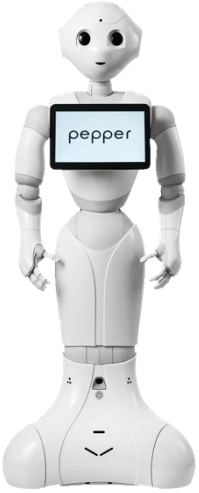
Speech Synthesis

Speech Synthesis Uses









Speech Synthesis

- Speech synthesis is the process of taking an input text and compositing a audio signal which emulates human speech, but which is built entirely from algorithmic / synthetic element. The model incorporates information on the correct pronunciation of text.
- Input Data = Written Text
  - Long texts
  - Conversational ‘turns’

Speech Synthesis: examples of the basic challenges

Heteronyms





bass

Blending of speech, specifically ‘liaison’

Vowel-consonant-vowel	
My name is Ann.	/maɪ neɪ **mɪ** zæn/
American accent	/əmɛrɪkə **n**æksənt/
LA	/ɛ **l**eɪ/

https://englishexplorations.check.uni-hamburg.de/4407/

Accents: Adding to the challenge



Accents are often best explained in terms of foreign languages, but the principles apply to the ‘accents’ which we grow up with.

https://youtu.be/8Wcqq0OKdyo?si=4mD4_Vc2nXPsbZOW



See the section in here on Irish and American accent.

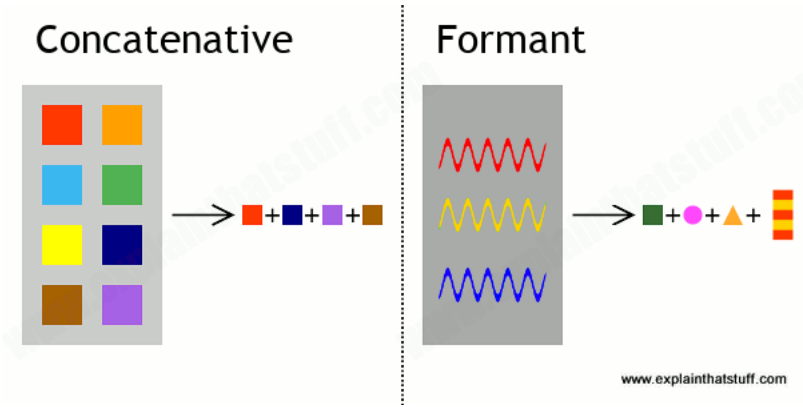
https://youtu.be/vpqjo1ea700?si=mDINo1sH13rvce9y

Speech Synthesis / TTS Types

See the Medium article.

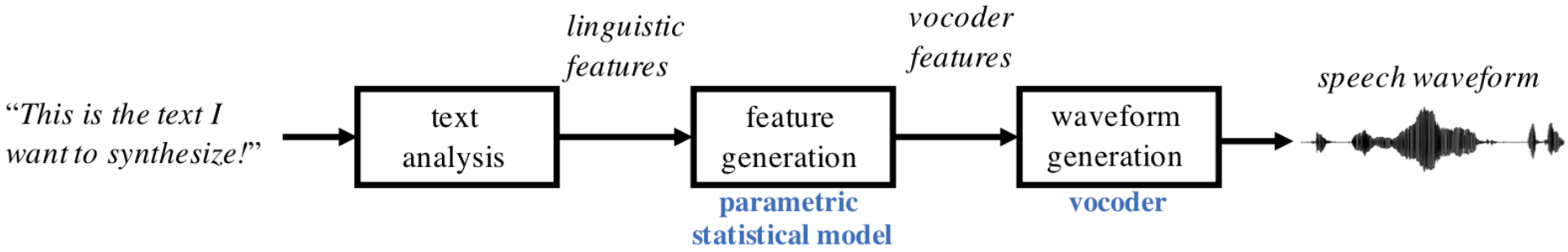
No need to memorize specific example systems, but the difference between types should be understood.

- Concatenative
- Formant Synthesis (rule-based)
- Parametric (Statistical)
- Hybrid / Deep Learning



Classic Speech Synthesis

Aside: It is useful to understand the basics of many classical NLT algorithms as modern algorithms are often achieving similar, but just hiding some of the decisions within neural network layers



https://wiki.aalto.fi/display/ITSP/Statistical+parametric+speech+synthesis

Text to Speech

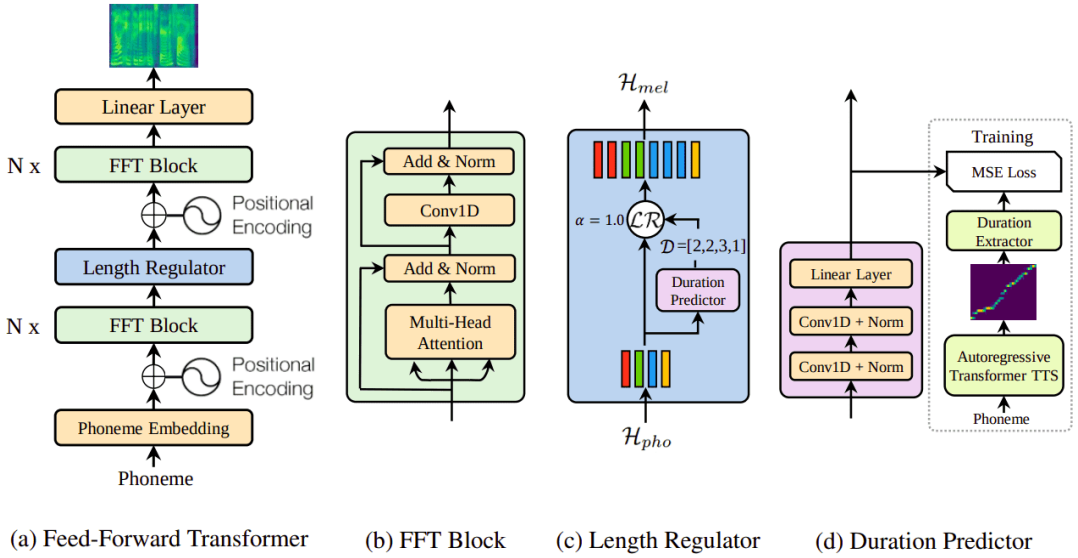
- Input Data = Written Text
  - Long texts
  - Conversational ‘turns’
- Classic Systems
  - Competent but not natural
- Modern Challenges
  - Accents
  - Unknown Words
  - Emotion
  - Emphasis (see also stress)
  - Personality Variation





Modern TTS can be Quite Complicated

(Detail not important here)



(a) Feed-Forward Transformer (b) FFT Block (c) Length Regulator (d) Duration Predictor

For deep learning based approaches, the model architectures can vary a lot, but the primary goal is the minimize the difference between produced spectrographs and target spectrographs for a given input sequence.

Text to Speech

- Input Data = Written Text
  - Long texts
  - Conversational ‘turns’
- Modern Challenges
  - Accents
  - Unknown Words
  - Emotion
  - Emphasis
  - Personality Variation
- State of the art
  - Amazon Polly
  - Elevenlabs TTS
  - Open Source: MeloTTS and Coqui TTS







Evaluating Text to Speech

- Quantitative evaluation of TTS is unusually difficult relative to other tasks
  - Focus on ‘quality’ of speech – a property similar to the goodness of music
- Traditional Human Based Evaluation
  - Mean Opinion Score
    - Score between 0 and 5
    - Human speech : 4.5 – 5
- Consider Also
  - Model size
  - Training cost
  - Fine-tuning ability
  - **Latency / production speed**
- Consider differences between
  - Intelligibility
  - Naturalness

Rating	Quality	Distortion
5	Excellent	Imperceptible
4	Good	Just perceptible, but not annoying
3	Fair	Perceptible and slightly annoying
2	Poor	Annoying, but not objectionable
1	Bad	Very annoying and objectionable

NaturalSpeech

- From Microsoft Asia
- See Samples
- https://speechresearch.github.io/naturalspeech/

Speech Synthesis Markup Language (SSML)

- A bit dated, but still widely supported across platforms
- XML Based
- Allows control of basic speech synthesis as well as prosody features
  - Pitch
  - Contour
  - Rate

```
<speak>
  <prosody rate="slow" pitch="-2st">Can you hear me now?</prosody>
  <prosody rate="fast" pitch="-2st">Can you hear me now?</prosody>
  <prosody rate="slow" pitch="-2st">Can you hear me now?</prosody>
  <prosody rate="slow" pitch="+5st">Can you hear me now?</prosody>
  <emphasis level="moderate">This is an important announcement</emphasis>
  This is an important announcement.
</speak>
```

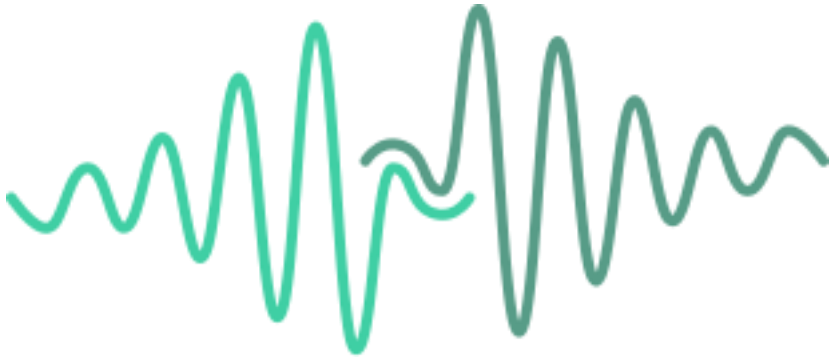
Custom Voices

- Neural Architectures easily allow fine-tuning of TTS systems with custom voice data.
  - This does not require full-from scratch model training
- Google at present offers this as a premium feature
  - Likely does not provide a perfect fix for Irish and other unsupported voices

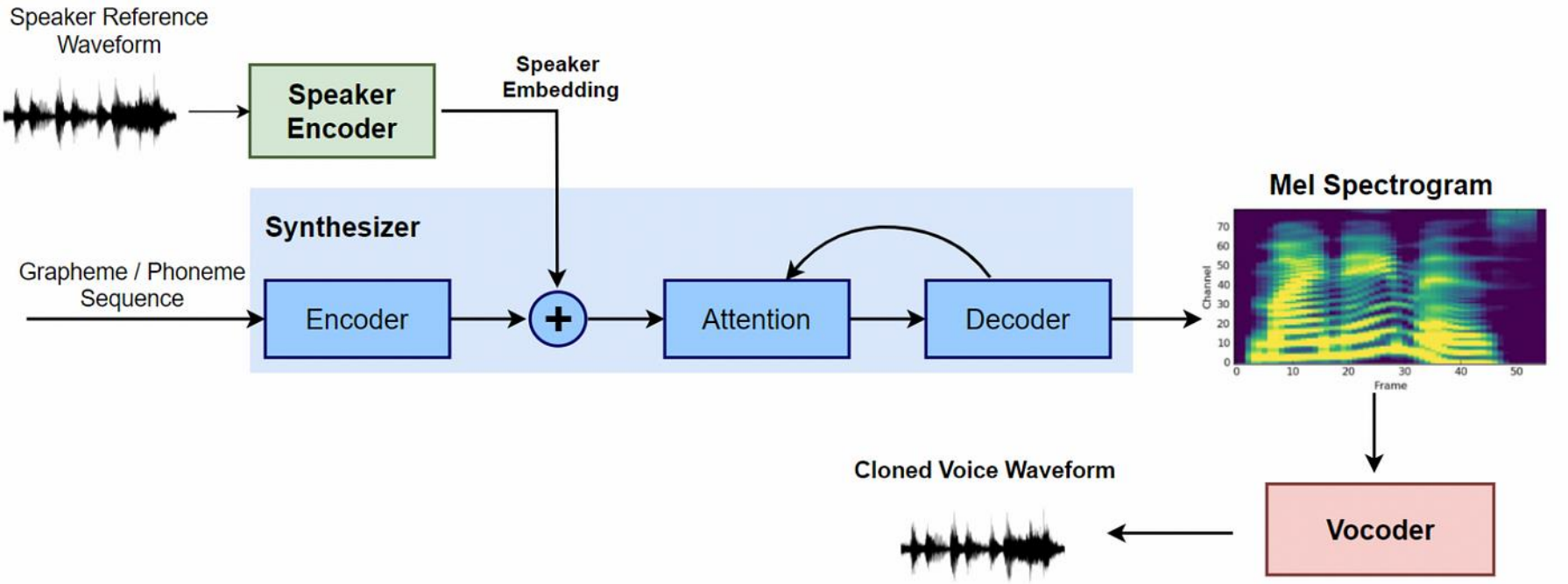
Voice Cloning

Voice Cloning

- Input Data = Written Text or Speech Data
- Output = Audio data with a target voice
- Challenges
  - Was very difficult until recently
  - Ethics
  - Getting it ”right” is challenging
  - Real time vs recorded
    - Real time factor (RTF) is an important metric. It should be below 1 for Real Time
- State of the art
  - See custom voices in the commercial TTS
  - Open Source: Tortoise TTS (transformer based)
  - OpenVoice V2



Voice Cloning - Overview



https://medium.com/mlearning-ai/voice-cloning-for-beginners-in-8-minutes-db046e009a70

Voice Cloning Ethics

- **Has this impacted you yet?**

Dealing with Cloned Voices

- **Anti-spoofing / deepfake detection:** separate classifiers are trained to tell real speech from synthetic speech (see the ASVspoof challenges). They tend to fail on generators they weren't trained on.
- **Watermarking:** some generators hide an inaudible watermark in their output, for example Resemble AI's PerTh in Chatterbox, Meta's AudioSeal and Google's SynthID. Qwen3-TTS and Kokoro don't. What are the limits of watermarking when anyone can run an open model without it?
- **Regulation:** the EU AI Act's transparency rules require AI-generated or manipulated audio ("deep fakes") to be disclosed or marked so machines can detect it. I haven't checked the exact dates these obligations apply from.

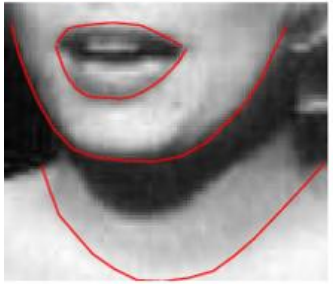
Visualizing Speech

Visemes

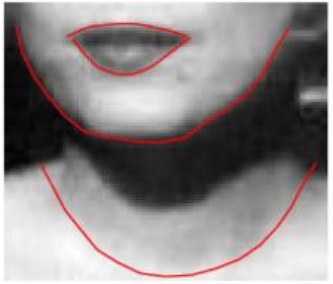
- Sets of speech sounds that look the same
  - Phonemes and visemes hence do not have a one-to-one relationship
- Descriptors of the visual arrangement of head and lips to sync with audio
- Can be output by a number of TTS tools
  - Azure Speech Service
  - Amazon Polly (request Speech Marks / visemes)

Visemes

A,AO,AW,AY



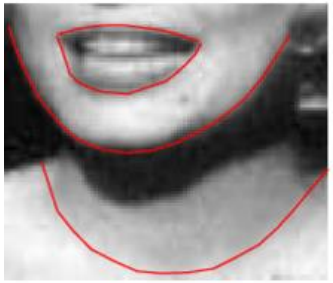
B,P,M



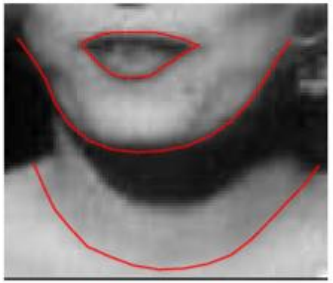
D,CH,JH,T,G,K,H,N



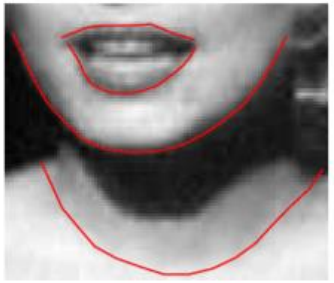
E,EL,EY,EH



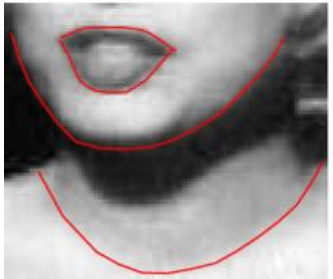
F,V



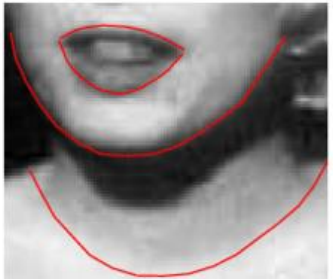
I,IY



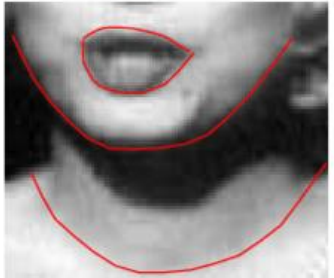
SH,S,Z



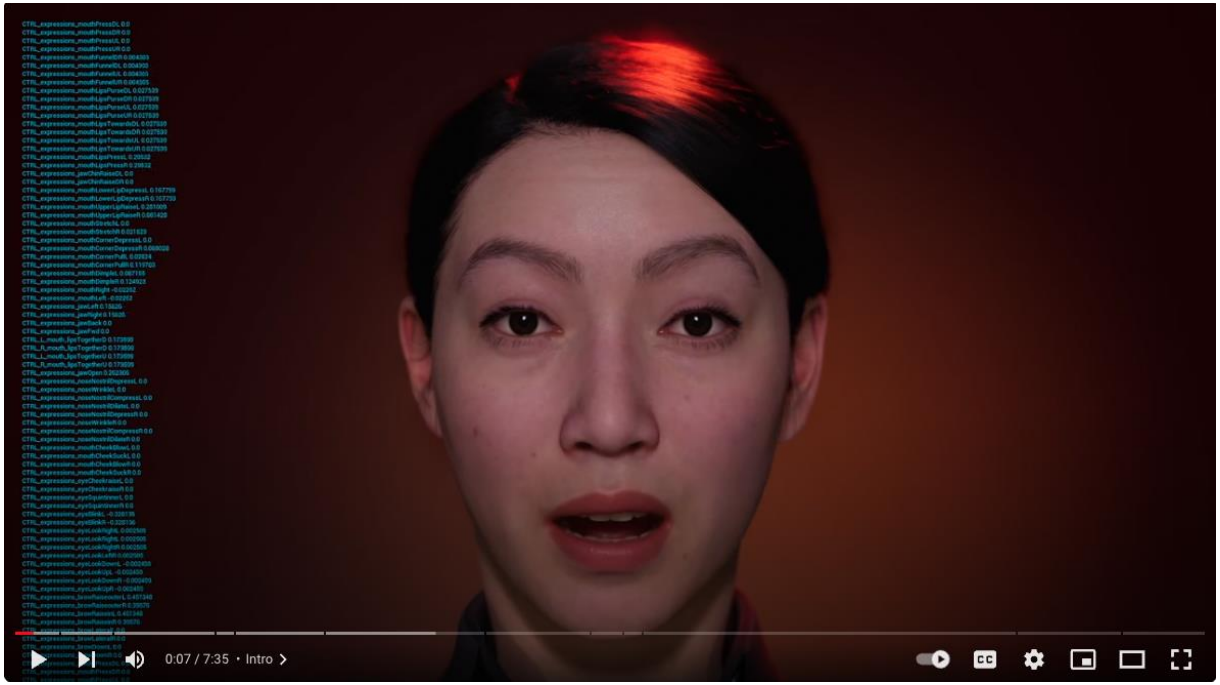
DH,TH



OW,OY,U,UW

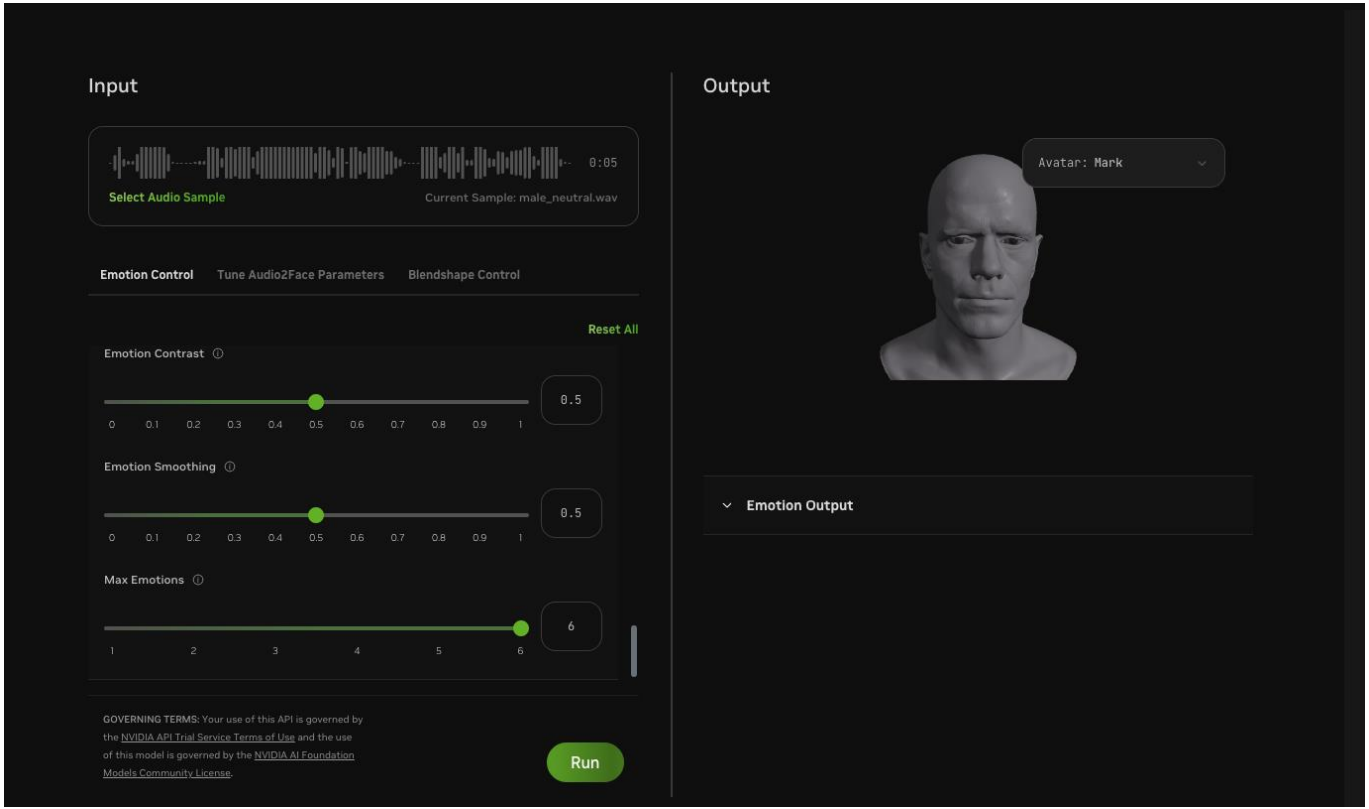


Still far from perfect



https://www.youtube.com/watch?app=desktop&v=y1wVykdmJNM

Nvidia Audio2Face



Challenges in Audio 2 Faces

- Real time need
- Solutions often based on game engines
- Microexpression Blending



Gestures in Communication

- Communication is more than sounds and lips
- Gestures using the hands and arms can convey significant information and help to keep communication focused
- Five Main Types of Gestures in Communication:
  - Symbolic
  - Diectic
  - Iconic
  - Pantamonic
  - Gesticulation / Beat Gestures

Research categorizes gestures into types that either replace or augment language.

Kendon's Continuum











LANGUAGE REPLACER — LANGUAGE AUGMENTER

SYMBOLIC

Gestures with a specific meaning that are consciously and subconsciously understood, e.g. symbol for "OK" and V for peace.

DEICTIC

Also known as pointing gestures, they help connect speech to an idea, object or location. e.g. "put that there"

ICONIC

Gestures that illustrate what is being said and add details to mental imagery. e.g. "this big", or "the plane flew like this"

PANTOMIMIC

Gestures that convey a narrative line without speech, e.g. "then we made a hard left turn"

GESTICULATION

Also known as beat gestures, they show emphasis, e.g. beating or tapping a finger

https://medium.com/@davidrose/the-age-of-gesture-876f92cdc947

Gesture in Avatars, e.g., Gesticulator





Challenges in Production of Non-Verbal Communication

- Accuracy
- Naturalness
- Uncanny Valley Effects
- ‘Alignment’

Side Note

McGurk Effect



https://www.youtube.com/watch?v=2k8fHR9jKVM

Lab Time